Fiscal Year:	FY 2025	**Task Last Updated:**	FY 11/05/2025
PI Name:	Opdensteinen, Patrick Ph.D.		
Project Title:	Streamlined Molecular Farming of VLP Therapeutics in Space		
Division Name:	Human Research		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:	TRISH--TRISH		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	(1) **Cancer**:Risk of Radiation Carcinogenesis (2) **Medical Conditions**:Risk of Adverse Health Outcomes and Decrements in Performance Due to Medical Conditions that occur in Mission, as well as Long Term Health Outcomes Due to Mission Exposures		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
PI Email:	popdensteinen@ucsd.edu	**Fax:**	FY
PI Organization Type:	UNIVERSITY	**Phone:**	760-783-7770
Organization Name:	University of California, San Diego		
PI Address 1:	9500 Gilman Ave		
PI Address 2:			
PI Web Page:			
City:	La Jolla	**State:**	AK
Zip Code:	92093	**Congressional District:**	50
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2024 TRISHRFA2401PD: Translational Research Institute for Space Health (TRISH) Announcement Soliciting Postdoctoral Fellowship Applications
Start Date:	09/01/2024	**End Date:**	08/31/2026
No. of Post Docs:	1	**No. of PhD Degrees:**	
No. of PhD Candidates:		**No. of Master' Degrees:**	
No. of Master's Candidates:		**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:		**Monitoring Center:**	TRISH
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Steinmetz, Nicole Ph.D. (MENTOR: University of California, San Diego)		
Grant/Contract No.:	NNX16AO69A-P0903		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	Drugs help to protect the health of astronauts during mission. Yet drugs expire faster in space compared to the ground. Refilling drugs in space is difficult because of long travel distances. The ability to produce drugs during missions would resolve these problems. Plants are the ideal platform for this task. First, plants can be sent into space as seeds to reduce transport costs. Second, plants can recycle air and water. Third, growing plants are good for the mental health of astronauts. Most importantly, plants can produce drugs under non-sterile conditions. All these properties are desirable to save resources. Plant-made drugs are usually harvested by breaking the plant tissue. This releases soluble and insoluble impurities. Removing these impurities is difficult in space, because there is no room for complex equipment. However, drugs can also be harvested from plants without breaking them. This is possible when producing drugs in the apoplast. We therefore propose a production process based on this cell compartment. The ideal model drugs for the production process are virus-like particles (VLPs). Plant VLPs resemble viruses, but are not infectious. VLPs can be used as vaccines or therapeutics. For example, the mentor lab established the application of VLPs as cancer immunotherapy. This could be applied to treat astronauts if they develop cancer from space radiation. We use modular DNA constructs for VLP production. This allows the researcher to adapt the proposed production process to other drugs (construct libraries). Libraries of DNA constructs will allow astronauts to respond to various medical problems (plug-and-play).
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	VLPs can be used as vaccines or therapeutics. For example, the mentor lab established the application of VLPs as cancer immunotherapy. This could be applied to treat astronauts if they develop cancer from space radiation.
Task Progress:	The first objective is to make the production process sustainable. Sustainable VLP production is achieved when the projected human dose (1 mg) can be produced on spacecrafts (1m² farming area). This objective involves the optimization of VLP production on the DNA and process level, and assessing the process resilience under simulated space conditions. The second objective is to establish the efficacy and safety of produced VLPs; this is a prerequisite for treating human patients. This objective involves the removal of impurities that can have adverse effects in humans using simple purification methods that can be adapted in space, as well as efficacy and safety validation. The third objective is to demonstrate plug-and play readiness of the production process by producing VLPs that contain custom RNA cargos. Plug-and-play readiness will facilitate the generation of construct libraries and enable astronauts to respond to emerging medical problems on demand. Key findings: The first key finding during this report period was that the model VLP-therapeutic cowpea mosaic virus (CPMV) accumulated in the apoplast without introducing genetic modifications. This allows us to omit Agrobacteria as delivery vector for modified CPMV. In consequence, the technology requirements of the process decreased because stirred tanks for cultivation of bacteria are no longer needed. Eluted CPMV was intact as confirmed by transmission electron microscopy. These results confirmed our hypothesis that intact VLP therapeutics can be harvested from plants without breaking them. The second key finding was that the accumulation of CPMV in plants increased under simulated space conditions. These results confirmed that the proposed production process is resilient during spaceflight. The third key finding during the first report period was that optimized infiltration buffers improved the CPMV purity and recovery in eluates. This finding confirmed our hypothesis that impurities such as plant proteins can be removed from CPMV preparations with simple methods. In the next funding period we will validate the efficacy and safety of CPMV preparations in vitro and in vivo and test plug-and play readiness of the production process.
Bibliography Type:	Description: (Last Updated: 06/17/2026)
Articles in Peer-reviewed Journals	Wu Z, Hsu SK, Opdensteinen P, Steinmetz NF. "A plant virus supramolecular NanoSponge for delivery of agricultural pesticides to the rhizosphere." ACS Appl. Mater. Interfaces. 2025 Sep 23. https://doi.org/10.1021/acsami.5c13714 , Sep-2025
Articles in Peer-reviewed Journals	Opdensteinen P, Affonso de Oliveira JF, Jung S, Steinmetz NF. "Toward translation of cowpea mosaic virus intratumoral immunotherapy with a scalable production process." Plant Biotechnology Journal. 2025 Nov;23(11):5013-31. https://doi.org/10.1111/pbi.70281 ; PMID: 40729524; PMCID: PMC12576442 , Nov-2025
Articles in Peer-reviewed Journals	Opdensteinen P, Caparco AA, Steinmetz NF. "Protein-based spherical nanoparticles for dsRNA delivery to nematodes-A platform technology for RNA silencing." Materials Today. 2025 Jun 11. https://doi.org/10.1016/j.mattod.2025.06.001 , Jun-2025
Articles in Peer-reviewed Journals	Karan S, Opdensteinen P, Ma Y, De Oliveira JFA, Steinmetz NF. "A replicon-based COVID-19 vaccine candidate delivered by tobacco mosaic virus-like particles." Vaccine. 2025 Apr 19;53:127063. https://doi.org/10.1016/j.vaccine.2025.127063 , Apr-2025
Awards	Opdensteinen P. "Borschers Badge of the RWTH Aachen University." Sep-2024
Awards	Opdensteinen P. "Third place in the 2025 Nanotechnology Entrepreneurship Challenge (NTEC)." Jun-2025